

PRE-DSA LOGIC BUILDING — QUESTION LIST

112 questions across 6 phases • @rohitTheMentor

PHASE 1 — CONDITIONAL THINKING

If-Else & Boolean Logic

L1 Simple Conditions (Getting Started)

1. Take a number and print whether it's positive, negative, or zero.
2. Check if a number is even or odd.
3. Check if a given year is a leap year.
4. Take a temperature value and print Cold / Warm / Hot using ranges.

L2 Nested If & Multiple Conditions

1. Take three sides and check if they form a valid triangle, and if so, its type.
2. Take marks (0-100) and print the corresponding grade.
3. Take the hour (0-23) and print Good Morning / Afternoon / Evening / Night.
4. Take a day number (1-7) and print the day name.

L3 Math & Number Logic

1. Take a 3-digit number and check if all digits are distinct.
2. Take (x, y) coordinates and determine the quadrant.
3. Check if an amount can be evenly divided into 2000/500/100 notes.
4. Check if a number is a perfect square (without using sqrt).

L4 Logical Operators & Compound Statements

1. Print Fizz / Buzz / FizzBuzz for divisibility by 3, 5, or both.
2. Take 24-hour time and print AM or PM.
3. Take a single digit (0-9) and print its word form.
4. Validate a password: length ≥ 8 and at least one digit.

L5 Creative / Tricky Logical Scenarios

1. Check if a point lies on the X-axis, Y-axis, or origin.
2. Check if three numbers can form a Pythagorean triplet.
3. Check if three numbers are in arithmetic progression.
4. Take a year and print its century.

PHASE 2 — LOOPING & PATTERNS

Iteration & Flow

L1 Basic Looping (Introductory)

1. Print the multiplication table of a given number.
2. Print the sum of the first n natural numbers.
3. Print the factorial of a given number.
4. Print the sum of all even numbers up to n.

L2 Number-based Looping Logic

1. Print the reverse of a given number.
2. Check if a number is a palindrome.
3. Check if a number is an Armstrong number.
4. Check if a number is prime or not.

L3 Mathematical & Logical Patterns

1. Print squares of numbers from 1 to n.
2. Find HCF (GCD) of two numbers using loops.
3. Find LCM of two numbers using loops.
4. Print all factors of a given number.

L5 Logical Loop Combinations

1. Check if a number's digit sum is even.
2. Count numbers between 1-500 divisible by 7 but not 5.
3. Find the smallest and largest digit in a number.
4. Print the sum of odd digits and even digits separately.

PHASE 3 — RECURSION

Thinking in Self-Reference

L1 Foundation of Recursion (Base + Recursive Case)

1. Print factorial of a number recursively.
2. Find the nth Fibonacci number recursively.
3. Print sum of first n natural numbers recursively.
4. Calculate power of a number (x^n) using recursion.

L2 Number-based Recursive Thinking

1. Reverse a number recursively.
2. Check if a number is a palindrome using recursion.
3. Find GCD of two numbers recursively (Euclid's algorithm).
4. Find the sum of digits of a number recursively.

L3 Pattern & Printing Problems

1. Print a triangle of stars recursively (top-down).
2. Print a triangle of stars recursively (bottom-up).
3. Print the multiplication table of n recursively.
4. Print a character pattern recursively (A, AB, ABC...).

L4 String-based Recursion

1. Reverse a string using recursion.
2. Check if a string is a palindrome using recursion.
3. Count vowels in a string recursively.
4. Remove all spaces from a string recursively.

PHASE 4 — BASIC ARRAYS

Iterative Logical Thinking

L1 Fundamentals of Arrays

1. Find the sum of all elements in an array.
2. Find the maximum element in an array.
3. Count how many elements are even and odd.
4. Print only elements greater than a given value k.

L2 Searching & Counting Logic

1. Check if a given element x exists in the array.
2. Count how many times a given element appears.
3. Find the first occurrence index of a given number.
4. Find the sum of even elements only.

L3 Transformation & Manipulation

1. Reverse an array without using built-in reverse.
2. Rotate an array by one position to the left.
3. Replace every negative number with 0.
4. Create a new array with squares of all numbers.

L4 Aggregate & Comparative Thinking

1. Find the common elements between two arrays.
2. Merge two arrays into a third array.
3. Find element-wise sum of two arrays.
4. Create a frequency array/map of numbers.

L5 Logical & Applied Array Problems

1. Find the second largest element.
2. Find the sum of all elements except the largest and smallest.
3. Count pairs of elements with sum equal to a given k.
4. Print all unique elements (occurring exactly once).

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PHASE 5 — STRINGS

Basic Logic Building

C1 Basic String Handling

1. Take a string and print its length.
2. Print the first and last character of a string.
3. Convert a string to uppercase.
4. Compare two strings lexicographically.

C2 Counting & Character Analysis

1. Count vowels and consonants in a string.
2. Find the frequency of each character (without a map).
3. Count how many spaces are in a sentence.
4. Print how many words start with a vowel.

C3 Reversing & Palindromic Thinking

1. Reverse a string without using built-in reverse.
2. Check whether a string is a palindrome.
3. Reverse each word in a sentence.
4. Remove the first and last character and print the rest.

C4 Character & Word Manipulation

1. Remove all vowels from a string.
2. Replace all vowels with '*'.
3. Remove duplicate characters from a string.
4. Swap case: uppercase to lowercase and vice versa.

C5 Word-level Thinking

1. Find the longest word in a sentence.
2. Capitalize the first letter of each word.
3. Convert the sentence to title case.
4. Count how many words have even length.

PHASE 6 — MIXED LOGICAL CHALLENGES

Applied Reasoning

C1 Number-Based Logical Combinations

1. Print all Armstrong numbers between 1 and 1000.
2. Print the reverse of a number.
3. Check if a number is palindrome.
4. Check if a number is perfect (sum of factors = number).

C2 String + Logic Mix

1. Check if two strings are anagrams (without using collections).
2. Check if two strings are rotations of each other.
3. Find the word with maximum vowels in a sentence.
4. Remove duplicate words from a sentence.

C3 Array + Looping Logic

1. Shift all zeros in an array to the end.
2. Find the second largest element in an array.
3. Rotate an array by one position to the right.
4. Find the sum of all elements at odd indices.

C4 Nested Logic & Pattern Flow

1. Print Pascal's triangle up to N rows.
2. Print all pairs in an array whose sum equals a given number.
3. Print a pattern of increasing characters (A, AB, ABC...).
4. Check if an array is sorted (ascending or descending).

C5 Applied Reasoning & Real-Life Logic

1. Given marks of students, find how many passed (≥ 40).
2. Validate a password (upper, lower, digit, special char).
3. Find common elements between two arrays.
4. Print all palindromic words from a sentence.

BONUS — PATTERN PRINTING (Striver's A2Z Sheet)

Must-do nested-loop patterns before DSA

1. Solid Square

```
OUTPUT
*****
*****
*****
*****
*****
*****
```

2. Left Triangle

```
OUTPUT
*
**
***
****
*****
```

3. Number Triangle

```
OUTPUT
1
12
123
1234
12345
```

4. Number-Repeat Triangle

```
OUTPUT
1
22
333
4444
55555
```

5. Inverted Triangle

```
OUTPUT
*****
****
***
**
*
```

6. Pyramid

```
OUTPUT
  *
 ***
*****
*****
*****
```